

Individual Contribution Report

Name	Index No	Course	Group No
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Fitting a Negative Binomial Regression Analysis for Dengue Incidence

Objective

This analysis models dengue incidence across districts using climatic and geographic predictors. The aim is to quantify how environmental conditions influence dengue transmission while accounting for population differences between districts.

Model Selection

Dengue cases are count data, which are commonly modeled using Poisson regression. However, Poisson models (even with log-transformed cases) remained **overdispersed**, where the variance exceeds the mean. To address this, a Negative Binomial Regression model was used. This model includes an additional dispersion parameter, allowing more flexible variance and producing more reliable statistical estimates.

A Generalized Linear Model (GLM) with a log link function was fitted to the dengue case counts. The expected number of cases was modeled as:

$$\ln\left(\frac{E[\text{Cases}]}{\text{Population}}\right) = \beta_0 + \beta_1(\text{Temp}) + \beta_2(\text{Rain}) + \beta_3(\text{Hum}) + \beta_4(\text{Elev}) + \text{District Effects}$$

Population was included as a log offset, allowing the model to estimate incidence rates rather than raw case counts. Climatic variables capture environmental conditions influencing mosquito populations, while elevation accounts for geographic differences in transmission suitability.

Model Output

Generalized Linear Model Regression Results										
				coef	std err	z	P> z	[0.025	0.975]	
Dep. Variable:	Cases	No. Observations:	899	Intercept	2.9328	1.795	1.634	0.102	-0.586	6.451
Model:	GLM	Df Residuals:	871	C(District)[T.Anuradhapura]	-0.2140	0.233	-0.917	0.359	-0.672	0.244
Model Family:	NegativeBinomial	Df Model:	27	C(District)[T.Badulla]	0.2501	0.188	1.327	0.184	-0.119	0.619
Link Function:	Log	Scale:	1.0000	C(District)[T.Batticaloa]	0.7298	0.240	3.042	0.002	0.260	1.200
Method:	IRLS	Log-likelihood:	-4930.1	C(District)[T.Colombo]	0.6510	0.244	2.672	0.008	0.174	1.128
Date:	Mon, 09 Mar 2026	Deviance:	1286.4	C(District)[T.Galle]	0.1608	0.245	0.656	0.512	-0.319	0.641
Time:	15:07:18	Pearson chi2:	1.38e+03	C(District)[T.Gampaha]	0.4066	0.243	1.671	0.095	-0.070	0.884
No. Iterations:	57	Pseudo R-squ. (CS):	0.6262	C(District)[T.Hambantota]	-1.0144	0.245	-4.133	0.000	-1.495	-0.533
Covariance Type:	nonrobust			C(District)[T.Jaffna]	1.1701	0.241	4.852	0.000	0.697	1.643
				C(District)[T.Kalutara]	-0.4000	0.244	-1.637	0.102	-0.879	0.079
				C(District)[T.Kandy]	0.8543	0.200	4.278	0.000	0.463	1.246
				C(District)[T.Kegalle]	0.1298	0.226	0.574	0.566	-0.313	0.573
				C(District)[T.Kilinochchi]	-1.6073	0.246	-6.546	0.000	-2.089	-1.126
				C(District)[T.Kurunegala]	-0.2981	0.233	-1.278	0.201	-0.755	0.159
				C(District)[T.Mannar]	-1.5911	0.246	-6.478	0.000	-2.072	-1.110
				C(District)[T.Matale]	0.0194	0.209	0.093	0.926	-0.390	0.429
				C(District)[T.Matara]	-0.1368	0.241	-0.567	0.571	-0.609	0.336
				C(District)[T.Moneragala]	-1.1607	0.230	-5.043	0.000	-1.612	-0.710
				C(District)[T.Mulativu]	-2.0837	0.250	-8.346	0.000	-2.573	-1.594
				C(District)[T.Nuwara Eliya]	-0.3270	0.110	-2.963	0.003	-0.543	-0.111
				C(District)[T.Polonnaruwa]	-0.7460	0.238	-3.129	0.002	-1.213	-0.279
				C(District)[T.Puttalam]	-0.7145	0.243	-2.938	0.003	-1.191	-0.238
				C(District)[T.Ratnapura]	0.3770	0.239	1.576	0.115	-0.092	0.846
				C(District)[T.Trincomalee]	0.1798	0.242	0.744	0.457	-0.294	0.653
				C(District)[T.Vavuniya]	-0.7498	0.233	-3.212	0.001	-1.207	-0.292
				Temp_avg	-0.4172	0.039	-10.682	0.000	-0.494	-0.341
				Precipitation_avg	-0.0168	0.011	-1.519	0.129	-0.038	0.005
				Humidity_avg	-0.0115	0.011	-1.044	0.295	-0.033	0.010
				Elevation	-0.0037	0.000	-13.095	0.000	-0.004	-0.003

Interpretation and Justification of Variables

The negative binomial regression model was chosen to account for overdispersion in the count of Cases, where the variance exceeds the mean. All predictors were included based on their theoretical relevance and their contribution to overall model fit.

Precipitation and humidity, although individually non-significant, were retained because their inclusion improves the model's statistical performance. Specifically, they contribute to a lower Akaike Information Criterion (AIC) and better overall goodness-of-fit, indicating that they help explain variation in case counts and reduce unexplained variability. Latitude and longitude were excluded, as including them did not meaningfully improve AIC or other model fit metrics, allowing for a simpler and more interpretable model without sacrificing predictive accuracy.

Overall, the model balances explanatory power, statistical robustness, and simplicity, providing a reliable framework for understanding the factors influencing case counts.

<https://github.com/deeps-pixel/dengue-forecasting-app.git>